

Relativity Set 1

Special Theory of Relativity Problems

1. Agent X is on a secret mission to the star α Centauri, 4.5 light years away. Enemy agent Y, desperate to stop him, manages to plant a time bomb, set to go off six months after take off, on X's interstellar ship. How fast must the ship travel so that X can land safely back home after a round trip (at α Centauri) and landing have no effect on the bomb? Assume that the (brief) periods of acceleration during take-off, turnaround (at α Centauri), and landing have no effect on the bomb.

2. A rocket based observer and his ground based twin set their respective clocks to zero at the instant they cross each other. Sometime later, the ground based twin observes the rocket through a powerful telescope. He finds that the rocket's clock is showing 4 o'clock, while his own clock (right next to him) is showing 5 o'clock.

What is the speed of the rocket (assumed uniform throughout) with respect to the Earth?

Hint: Do not forget the time it takes for light to travel between the rocket and the Earth.

3. Alice has two clocks — one at her origin and another 8 light-hours away — which she has synchronized. Bob rushes past her (in the direction of the second clock) on a rocket ship at a speed of $0.8c$ at the moment her clock is showing 0, and sets his own clock to 0 at that instant.

What do Bob's and Alice's clocks show when he is just beside the second clock? From these clock readings, calculate the speed of Alice's clocks as observed by Bob.

Explain how Bob can reach the conclusion that Alice's clocks are running slow compared to his own — despite your answer to the last part!

4. A train of proper length L_0 is approaching a station at a speed of βc . Light signals emitted by the front and back ends of the train reach an observer on the station simultaneously.

Calculate the distance between the points of emission of the light signals, as measured by the stationary observer.

This problem should illustrate the difference between the "observed length contraction" and what a moving object actually looks like!
